The emergence of numerical representations in communicating artificial agents

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Human languages provide efficient systems for expressing numerosities, but whether the sheer pressure to communicate is enough for numerical representations to arise in artificial agents, and whether the emergent codes resemble human numerals at all, remains an open question. We study two neural network-based agents that must communicate numerosities in a referential game using either discrete tokens or continuous sketches, thus exploring both symbolic and iconic representations. Without any pre-defined numeric concepts, the agents achieve high in-distribution communication accuracy in both communication channels and converge on high-precision symbol–meaning mappings. However, the emergent code is non-compositional: the agents fail to derive systematic messages for unseen numerosities, typically reusing the symbol of the highest trained numerosity (discrete), or collapsing extrapolated values onto a single sketch (continuous). We conclude that the communication pressure alone suffices for precise transmission of learned numerosities, but additional pressures are needed to yield compositional codes and generalisation abilities.

1. Communicating and representing numerosities

Number words, such as *two, ten, sixteen*, allow humans to encode any numerosity up to infinity, relying on finite memory. Their encoding efficiency lies in several properties. First, they exhibit **precision**: each word maps one-to-one onto a numerosity; this mapping can be extended indefinitely through extrapolation, assigning a new word to each added element. They also follow transparent **compositional rules**, reducing the number of forms to memorise: for example, knowing the words for units and tens allows speakers to interpolate intermediate numerosities. These properties enable learnability, making concept–word mappings easy to infer and **generalise** to novel words and novel numerosities: in humans, understanding the structure of number words unlocks precise representation of any numerical concept (Barner, 2017), whereas without number words, quantity representation has been argued to be only approximate. Finally, the form of number words is **arbitrary**, as no property of the word suggests the numerosity they refer to; however, as commonly in language, most frequently occurring number words are shorter (Pajot, Sablé-Meyer, & Dehaene, 2025). However, written representations of numbers vary in arbitrariness and can start as iconic representations,

where the visual properties of number symbols directly reproduce the numerosity to encode (Overmann, 2025).

Although numerical systems emerged independently in different cultures (Coolidge & Overmann, 2012), number words across languages largely share these properties (Comrie, 2013), suggesting that they reflect optimisation to similar encoding pressures (Xu, Liu, & Regier, 2020; Denić & Szymanik, 2024).

Language evolution theories propose that structured systems arise under pressures for transmission, economy, and generalisation (Kirby, Cornish, & Smith, 2008). Across cultures, numbers originated as a cultural invention for unambiguous communication, used to record stored goods and track quantities in transactions (Menninger, 2013). We therefore ask whether communicative pressures shape number words and, in turn, enable the representation of number concepts.

Specifically, we ask (i) whether the properties observed in natural language numerals also arise in a code self-optimised through communication, (ii) which aspects of communication support them and (iii) whether the resulting code supports generalisation to unseen numerosities, bootstrapping their conceptual representation. To this end, we implement a referential game (Lewis, 1969; Boldt & Mortensen, 2024; Brandizzi, 2023), in which a communication protocol is shaped solely through collaborative interaction between two neural-network agents without prior knowledge of numerals. We test whether the pressure to communicate is sufficient for agents to develop a code, and whether it shows precision, compositionality, generalisation, and arbitrariness as emerging properties. We also assess whether frequency and learning strategy (interpolation vs extrapolation) affect communication success and code structure.

2. Environmental Setup

Communication Task. We used a referential game setup (Fig. 1a), where two neural-network agents communicate about numerosities. In each trial, a sender agent is presented with an image displaying a certain number of dots and sends a message to a receiver agent. The receiver, based on the message, must identify the *target* image showing the same number of dots, among distractors with different numerosities. Upon target selection, both agents receive feedback; in this way, the communication protocol emerges while solving the task. The setup has a *training* phase, where agents receive feedback and therefore develop their code, and a *generalisation* phase, where they communicate about novel numerosities.

Stimuli. We designed the stimuli to ensure that agents rely only on numerosity, preventing confounds of other visual properties, like shape, colour, area or perimeter occupied by the dots. Stimuli images are 256 x 256 pixels in size, each depicting a different number of black dots on a white background (e.g. Fig. 1b). In order to make information on numerosity independent of other visual variables such as the area occupied by black pixels, we constrain the total black area occupied by dots to 5-10% of the canvas for all numerosities.

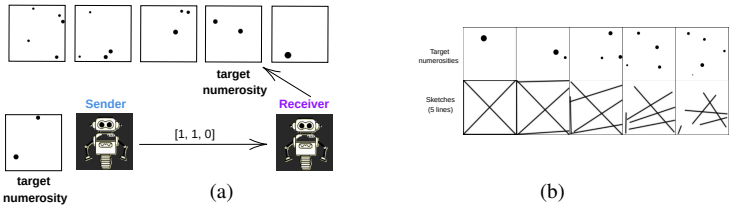


Figure 1. (a) A referential game: the Receiver has to guess, based on a Sender's discrete message, the correct target out of 5 possible numerosities. Notice that although the target is also 2, it is not the same instance as seen by the Sender. (b) Examples of target numerosities $\{1, 2, 3, 4, 5\}$ and their 5-line corresponding sketches.

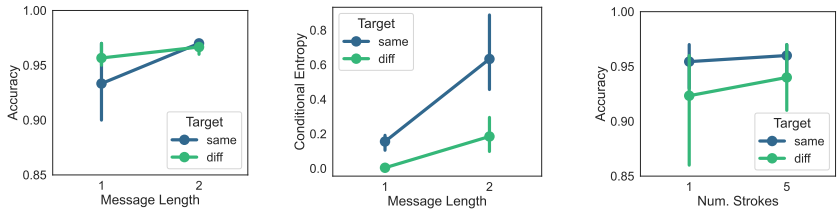
Conditions. For the purpose of this work, the main training set is mostly limited to numerosities between 1 and 5 (although we experimented with larger training sets, such as 1 to 20, and observed similar results). The extrapolation/interpolation evaluation can include one other numerosity. The exact training and testing numerosities are specified for each experiment. To test if representational abilities of numerosity emerge independently of the low-level visual properties, we test two target setups, *Same* and *Diff*, which indicate whether the agents have the exact *same* target image, or two *different* instances from the same numerosity class.

Agents. Both agents use a shared pretrained ViT encoder for processing the dot images (Radford et al., 2021). We model two types of communication channels. In the **discrete** setup, the sender is equipped with an LSTM encoder (Hochreiter & Schmidhuber, 1997), which outputs a message as a sequence of discrete tokens chosen over a predefined vocabulary; the receiver's LSTM reads the message and accordingly selects an image as a possible target. In the **continuous** setup, the sender draws a fixed number of straight lines on a blank canvas, i.e. a sketch (Mihai & Hare, 2021). The receiver reads the sketch through the same ViT encoder used for the candidate target images. A multi-class hinge loss is employed to maximise the probability that the correct target image is selected. Training happens on a set of numerosities as specified per experiment, and the testing is both in-range and on unseen values.

Metrics. We assess communication success via task accuracy, defined as the proportion of trials in which the receiver selects the numerosity presented to the sender. Additionally, we assess the properties of the emerging communication code: *Efficiency* via message length (tokens) or sketch strokes/span; *Precision* as conditional entropy of numerosity given message (sketches are discretised via clustering); *Generalisation* by accuracy on unseen numerosities and where their messages map among seen classes; and *Arbitrariness* by skewing training frequencies and checking if more frequent numbers get shorter or simpler codes.

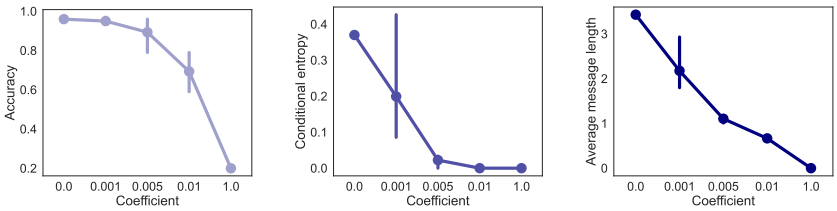
3. Results

Communication success and code precision First, we observe that agents can reliably communicate trained numerosities, even in the *Diff* condition, where the sender and receiver see different instances of the same numerosity (Fig. 2a and 2c). The system becomes more precise in the different setups indicated by the lower entropy (Fig. 2b and 4a). Under length regularisation (Fig. 3), we observe that messages become shorter with minimal accuracy loss and lower entropy, indicating the pressure toward efficient bijective encodings. This shows that even without language-internal conventions, precision emerges from the joint pressure of coordination and information bottleneck.



(a) (b) (c)

Figure 2. (a) Discrete communication accuracy and (b) message entropy, and (c) continuous (i.e. sketch-based) communication accuracy for setups involving identical target pictures (same - blue line) and different pictures showing the same numerosity (diff - green line). Model trained and tested on numerosities 1 to 5. Results averaged over 3 seeds.



(a) (b) (c)

Figure 3. Discrete Communication - Agent messaging behaviour when longer messages are penalised by regularising the game loss. Model trained on numerosities $\{1, 2, 3, 4, 5\}$, with a variable message length allowed up to **5** tokens, and $vocabulary = 3$. (a) Accuracy (b) conditional entropy of the mapping, and (c) average message length for different regularisation coefficients.

Arbitrariness outweighs frequency We next vary the training context to assess whether more frequently occurring numerosities are encoded more economically compared to the least frequent ones. We find that manipulating class frequency (Uniform/Increase/Decrease setups detailed in supplementary) barely changes accuracy (Fig. 4b) and does not induce shorter codes for frequent numerosities. The sketch channel tends to scale with the spatial span of the input image elements, rather than with frequency (Fig. 4c). Human numerals often show mixed arbi-

trariness with some high-frequency lexicalisation. In this set of experiments, the absence of transmission costs beyond accuracy likely weakens Zipf-like simplification, and the arbitrariness survives because it is *good enough* for coordination.

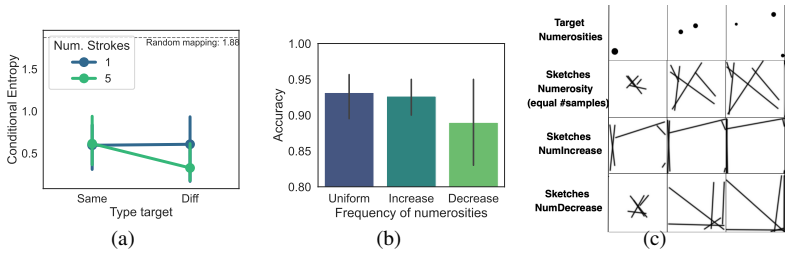


Figure 4. (a) Conditional entropy for communication using 1 and 5-line sketches with two target contexts (same or different between the 2 agents). (b) Communication accuracy on two altered numerosity datasets with an unequal number of training samples: increasing from class 1, having the fewest samples, to class 5, having the most, and vice versa. (c) Test sketches from training on the 3 different datasets: equal, increasing and decreasing number of class samples.

Generalisation and compositionality We further investigate whether the code developed by the agents can generalise to unseen numerosities (Discrete setup shown in Fig. 5 Top). We find that under out-of-distribution testing, agents typically reuse the message or symbol for the largest trained numerosity. The interpolation analysis, testing within-the-range numerosities, also indicates a lack of systematic structure. Sketch-based communication behaves similarly, although on average, extrapolations with a continuous channel are better than chance. However, agents do not develop a compositional sketch but rather map everything larger than in training onto a single sketch, which usually resembles the highest seen numerosity (*e.g.*Fig. 5-Bottom, and supplementary). The emerging signals stabilise as holistic labels, with a separate message/sketch for each trained numerosity and no subparts are reused to compose new messages.

4. Implications for Language Evolution

Coordination yields precision and numerical representations. The agents are able to quickly learn low-entropy codes, precisely communicating about trained numerosities. Numerosities are communicated independently of visual features, indicating the acquisition of abstract numerical representations.

Iconicity helps, but arbitrariness wins. The sketch communication channel shows some size and span cues, with sketch size capturing the spread of the dots, rather than the quantity; but both discrete and continuous communication channels settle into mostly arbitrary, class-specific signals.

Frequency alone doesn't yield efficient codes. Human languages often make frequent words shorter, but here, without explicit costs for longer messages,

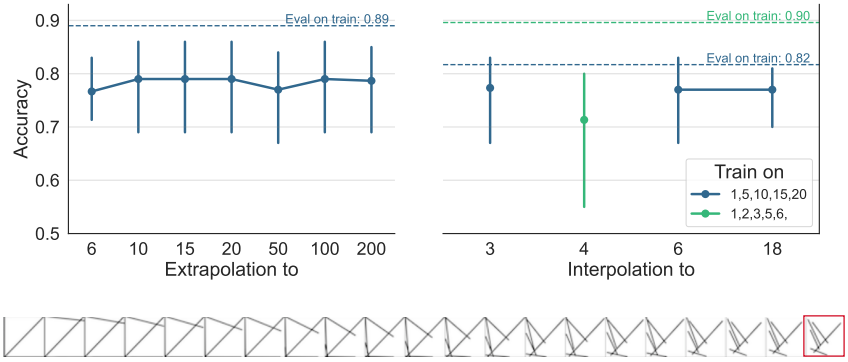


Figure 5. (Left) **Discrete extrapolation**: model trained on $\{1, 2, 3, 4, 5\}$. Test communication accuracy is reported for each test set (train + one out-of-distribution numerosity). The model uses the same message that was previously chosen for the highest pretrained numerosity (*e.g.*5) for every new higher extrapolation. (Right) **Interpolation** results from two different training sets. (Bottom) Example of sketch extrapolations from training within $\{1 - 20\}$ to *25* highlighted in red.

skewed input distributions do not result in more efficient codes. Adding production and perception costs could aid the brevity of the emergent codes.

Lack of compositionality, lack of generalisation . When faced with higher, untrained numerosities, agents reuse the message associated with the largest trained numerosity. This limitation may relate to the lack of compositionality in the emergent communication system: agents use holistic messages, and neither reuse interpretable message parts nor develop rule-based compositions. While holistic messages encoding non-decomposable bundles of features commonly emerge in referential games (LaCroix, 2025; Freeborn, 2025), compositionality plays a crucial role in bootstrapping numerical representations, as evidenced by human learning. Inducing such compositionality in this context remains a key challenge. Unlike referents defined by multiple features (*e.g.*, colour and shape), numbers vary along a single ordered dimension, offering no inherently decomposable features to encode. Successful generalisation, therefore, requires the emergence of an arbitrary but systematic partitioning of this dimension, rather than the transparent mapping of perceptual features. How such a structure can be induced remains an open question. Alternative learning setups may help, such as population training (e.g. Mahaut et al., 2025) or other mechanisms that promote compositionality in both human communication and neural networks, such as iterated learning, vocabulary constraints, and biases toward symbol reuse (Lake & Baroni, 2023; Kirby et al., 2008). Exploring such mechanisms may therefore be key to enabling systematic numerical generalisation.

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